MISR UNIVERSITY
FOR SCIENCE & TECHNOLOGY
College of Information Technology



جامعة مصر
للعلوم والتكنولوجيا
كلية تكنولوجيا المعلومات

CS481- Pattern Recognition

Course Instructor

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“Education is the most powerful weapon which you can use to change the world.”

— Nelson Mandela

Expected Code of Conduct











Marginal probability

- **Marginal probability** is the probability of an event irrespective of another event.
- The probability of a specific value of one input variable is the marginal probability across the values of the other input variables.

		Smoker?		
		Yes	No	Total
Gender	Male	.41	.07	.48
	Female	.47	.05	.52
	Total	.88		1.00

- P(X=Male) = sum P(X=Male, Y=yi) for all i

Joint probability

INDEPENDENT EVENTS

The outcome of one event does not affect the outcome of the other event.

$$P(A \cap B) = P(A) \times P(B)$$

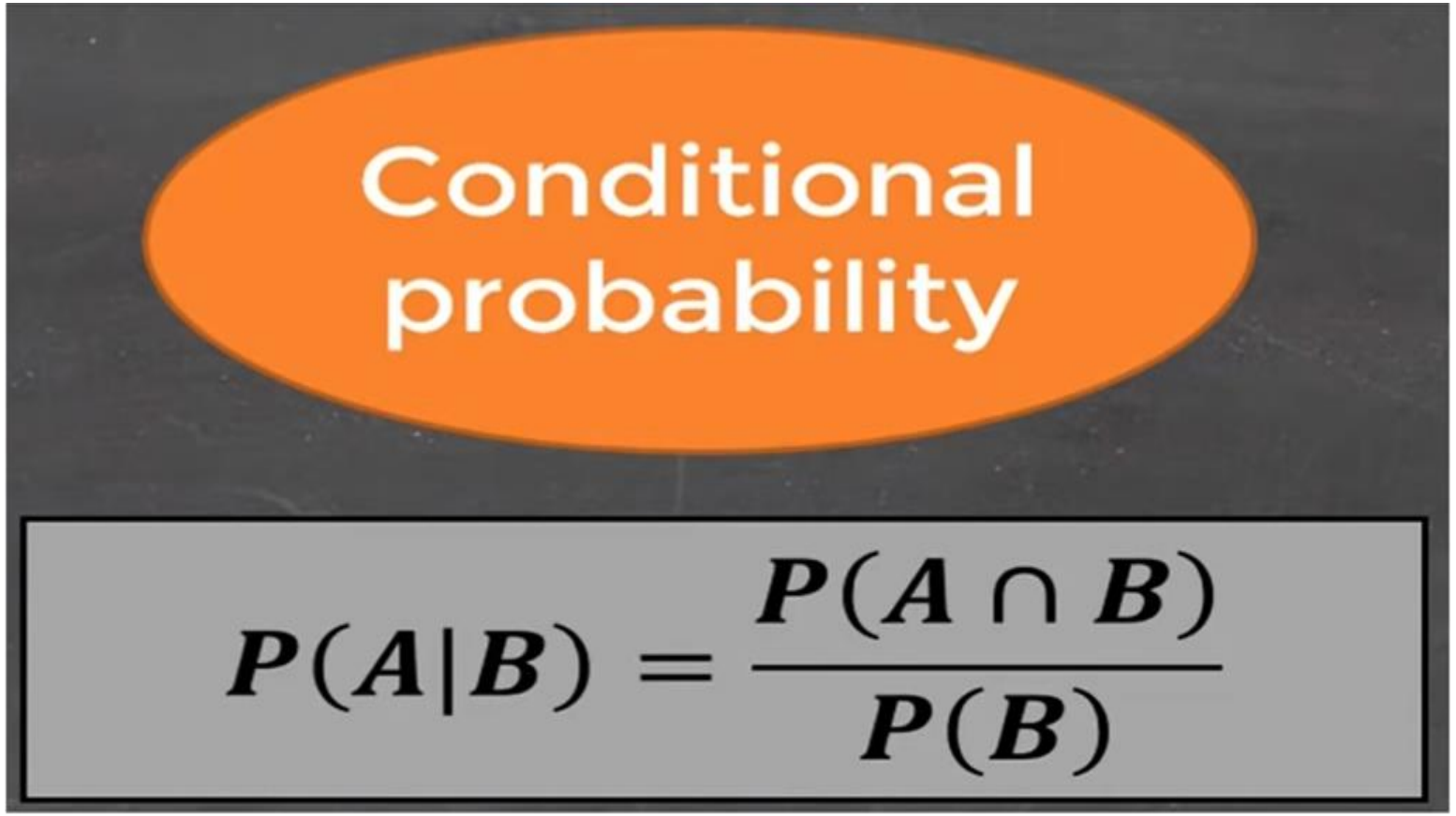
DEPENDENT EVENTS

The outcome of one event does influence the outcome of the other event.

$$P(A \cap B) = P(A) \times P(B \text{ after } A)$$

Conditional Probability

- **Conditional probability equals joint probability over marginal probability.**



Bayes Classifier

- Since
  - $P(A \cap B) = P(B \cap A)$,
  - $P(A \cap B) = P(A) * P(B|A)$
  - $P(B \cap A) = P(B) * P(A|B)$

  Equal
- So,
  - $P(A)*P(B|A) = P(B)*P(A|B)$,
  - $P(A|B) = [P(A)*P(B|A)] / [P(B)]$
- We refer to
  - $P(A|B)$ as posterior probability,
  - $P(A)$ as prior probability
  - $P(B|A)$ as likelihood.
  - $P(B)$ as marginal probability

Example

Problem statement

- A new medical test is used to detect whether a patient has a certain cancer or not, whose test result is either + (*positive*) or - (*negative*)
- For patient with this cancer, the probability of returning a *positive* test result is 0.98 i.e $p(+|cancer)=0.98$
- For patient without this cancer, the probability of returning a *negative* test result is 0.97 i.e $p(-|nocancer)=0.97$
- The probability for any person to have this cancer is 0.008
  i.e $p(cancer)=0.008$

Question

If positive test result is returned for some person, does he/she have this kind of cancer or not? P(cancer|positive) or P(nocancer|positive) i.e $(p(w_1|+)$ *or* $p(w_2|+))$

Example 2

ω_1 : cancer $\quad$ ω_2 : no cancer $\quad$ $x \in \{+, -\}$

$$P(\omega_1) = 0.008 \qquad P(\omega_2) = 1 - P(\omega_1) = 0.992$$

$$P(+ \mid \omega_1) = 0.98 \rightarrow P(- \mid \omega_1) = 1 - P(+ \mid \omega_1) = 0.02$$

$$P(- \mid \omega_2) = 0.97 \qquad P(+ \mid \omega_2) = 1 - P(- \mid \omega_2) = 0.03$$

$$P(\omega_1 \mid +) = \frac{P(\omega_1)P(+ \mid \omega_1)}{P(+)} = \frac{P(\omega_1)P(+ \mid \omega_1)}{P(\omega_1)P(+ \mid \omega_1) + P(\omega_2)P(+ \mid \omega_2)}$$

$$= \frac{0.008 \times 0.98}{0.008 \times 0.98 + 0.992 \times 0.03} = 0.2085$$

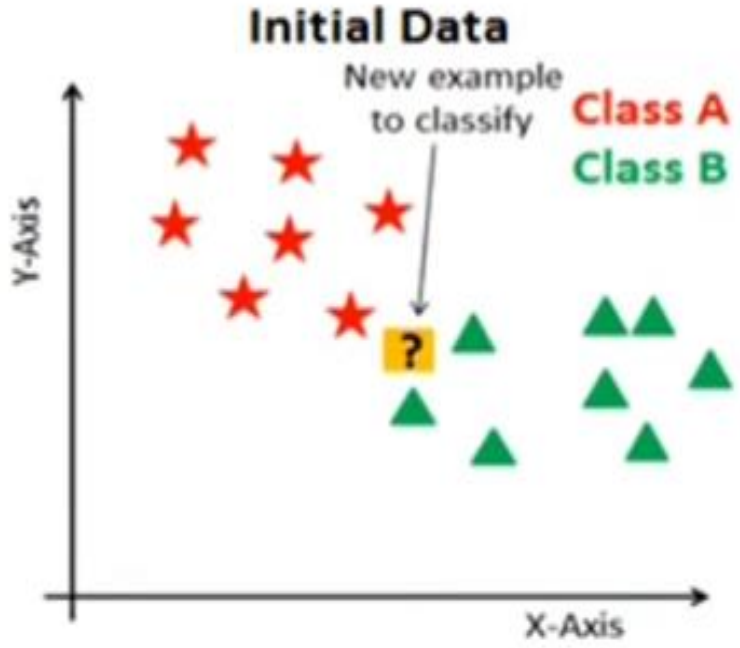
$$P(\omega_2 \mid +) = 1 - P(\omega_1 \mid +) = 0.7915$$

Naive Bayes' Classifier

- The occurrence of two events, where they are dependent events is called conditional probability: P(A|B).
- Sometimes, many event occurs while one of them depends of the rest of events: P(A|B,C,D,E, F,..).
- Bayes' classifier concerns with the dependency of an event on only one event: P(A|B).
- Naïve Bayes' classifier concerns with the dependency of an event on many events: P(A|B,C,D,E,F,...)

Naive Bayes' Classifier

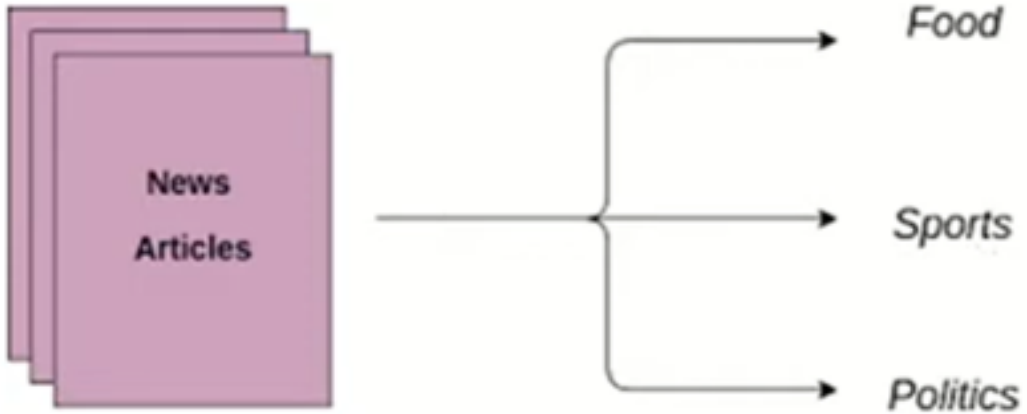
- Naïve Bayes' classifier is the most straight forward algorithm which is suitable for large chunk of data.
- It uses the Bayes' theorem of probability for predicting or classifying unknown instance.
- Naïve Bayes' classifier works on a set of features. Features are the characteristics or attributes that affects the results of classification.



Naive Bayes' Classifier

- ❑Text Classification
- ❑Sentiment analysis
- ❑ Spam Detection







Naive Bayes' Classifier

- **Naïve Bayes Formula:**

$$P(Y = yes \mid X) = \frac{P(X \mid Y) \cdot P(Y)}{P(X)}$$

- Where:
  - **X** = Feature vector (x_1, x_2, ..., x_m)
  - **Y** = Class label
- **Expanded form:**

$$P(Y = yes \mid X) = \frac{P(x_1 \mid Y) \cdot P(x_2 \mid Y) \cdots P(x_m \mid Y) \cdot P(Y)}{P(x_1) \cdot P(x_2) \cdots P(x_m)}$$

Naive Bayes' Classifier

- **Full Bayes Formula:**

$$P(y|x_1, x_2, \ldots xm) = \frac{P(y)\prod_{i=1}^{m} P(x_i \mid y)}{P(x_1)P(x_2)P(x_3)\cdots P(x_m)}$$

- **Simplified (Proportional) Form:**

$$P(y|x_1, x_2, \ldots xm) \propto P(y)\prod_{i=1}^{m} P(x_i \mid y)$$

- **Classification Rule (in red):**

$$y = \arg\max_{y} \; P(y)\prod_{i=1}^{m} P(x_i \mid y)$$

Naive Bayes' Example - Weather Dataset

Outlook	Temp	Humidity	Windy	Play
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

Naive Bayes' Example - Weather Dataset

Feature		Play or Not?	
		Yes	**No**
Outlook	Sunny	3	3
	Overcast	**4**	**0**
	Rainy	2	2
Temperature	Hot	2	2
	Mild	4	2
	Cool	3	1
Humidity	High	3	4
	Normal	6	1
Windy	True	3	3
	False	6	2
Probability		**9/14**	**5/14**

Outlook	**Temp**	**Humidity**	**Windy**	**Play**
Rainy	Hot	Normal	True	**?**

Naive Bayes' Classifier

- **Bayes Theorem Formula:**

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

- **Simplified (Proportional) Form:**

- Let E represent the evidence (the weather conditions):

E={Outlook = rainy, Temp = Hot, Humidity = Normal, Windy = True}

The two equations in the image calculate

P(Play = Yes |E) and P(Play = No |E):

Naive Bayes' Classifier

- **Probability of "Play = Yes" given the Evidence**

$$P(\textbf{Play = Yes} \mid \textbf{E}) = \frac{P(E \mid Play = Yes) * P(\textbf{Play = Yes})}{P(E)}$$

❑**Numerator:** $$P\left(\begin{array}{l}\text{outlook} = \text{rainy},\\ \text{Temp} = \text{Hot},\\ \text{Humidity} = \text{Normal},\\ \text{Windy} = \text{True}\end{array}\middle|\ \text{Play} = \text{Yes}\right) \times P(\text{Play} = \text{Yes})$$

❑**Denominator**: $$P\left(\begin{array}{l}\text{outlook} = \text{rainy},\\ \text{Temp} = \text{Hot},\\ \text{Humidity} = \text{Normal},\\ \text{Windy} = \text{True}\end{array}\right)$$

Naive Bayes' Classifier

- Probability of "Play = Yes" given the Evidence

$$P(\text{Play = Yes } |\text{E}) = \frac{P(E|Play = Yes) * P(\text{Play = Yes})}{P(E)}$$

- Probability of "Play = No" given the Evidence

$$P(\text{Play = No } |\text{E}) = \frac{P(E|Play = No) * P(\text{Play = No})}{P(E)}$$

Naive Bayes' Classifier

$$P\left(Play = Yes \,\middle|\, \begin{matrix} rainy, \\ Hot, \\ Normal, \\ True \end{matrix}\right) = P\left(\begin{matrix} rainy, \\ Hot, \\ Normal, \\ True, \end{matrix} \,\middle|\, Yes\right) \times P(Yes)$$

$$P\left(Play = Yes \,\middle|\, \begin{matrix} rainy, \\ Hot, \\ Normal, \\ True \end{matrix}\right) = P(Rainy|Yes) \times P(Hot|Yes) \times P(Normal|Yes) \times P(True|Yes) \times P(Yes)$$

$$P\left(Play = Yes \,\middle|\, \begin{matrix} rainy, \\ Hot, \\ Normal, \\ True \end{matrix}\right) = \frac{2}{9} \times \frac{2}{9} \times \frac{6}{9} \times \frac{3}{9} \times \frac{9}{14}$$

Naive Bayes' Classifier

- **Probability of "Play = No" given the Evidence**

$$P(\text{Play = No} \mid \text{E}) = \frac{P(E \mid Play = No) * P(\text{Play = No})}{P(E)}$$

❑**Numerator:** $$P\left(\begin{array}{l}\text{outlook} = \text{rainy},\\ \text{Temp} = \text{Hot},\\ \text{Humidity} = \text{Normal},\\ \text{Windy} = \text{True}\end{array}\middle|\ \text{Play} = \text{No}\right) \times P(\text{Play} = \text{No})$$

❑**Denominator**: $$P\left(\begin{array}{l}\text{outlook} = \text{rainy},\\ \text{Temp} = \text{Hot},\\ \text{Humidity} = \text{Normal},\\ \text{Windy} = \text{True}\end{array}\right)$$

Naive Bayes' Classifier

$$P\left(Play = \text{No} \,\middle|\, \begin{matrix} rainy, \\ Hot, \\ Normal, \\ True \end{matrix}\right) = P\left(\begin{matrix} rainy, \\ Hot, \\ Normal, \\ True, \end{matrix} \,\middle|\, \text{No}\right) \times P(\text{No})$$

$$P\left(Play = \text{No} \,\middle|\, \begin{matrix} rainy, \\ Hot, \\ Normal, \\ True \end{matrix}\right) = P(Rainy|\,\text{No}) \times P(Hot|\,\text{No}) \times P(Normal|\,\text{No}) \times P(True|\,\text{No}) \times P(\text{No})$$

$$P\left(Play = \text{No} \,\middle|\, \begin{matrix} rainy, \\ Hot, \\ Normal, \\ True \end{matrix}\right) = \frac{2}{5} \times \frac{2}{5} \times \frac{6}{5} \times \frac{1}{5} \times \frac{5}{14}$$

Laplace Correction

- The value of *P(Overcast |No) = 0*, there were no overcast where we play
- *It makes sense*, but when we have a model with many features, *the entire probability will become zero* because one of the feature's value was zero
- To avoid this, we *increase the count of the variable with zero to a small value (usually 1) in the numerator*, so that the overall probability does not become zero
- The correction is called "*Laplace Correction*".
- Most *Naïve Bayes* model implementation *accept this*



Thank You